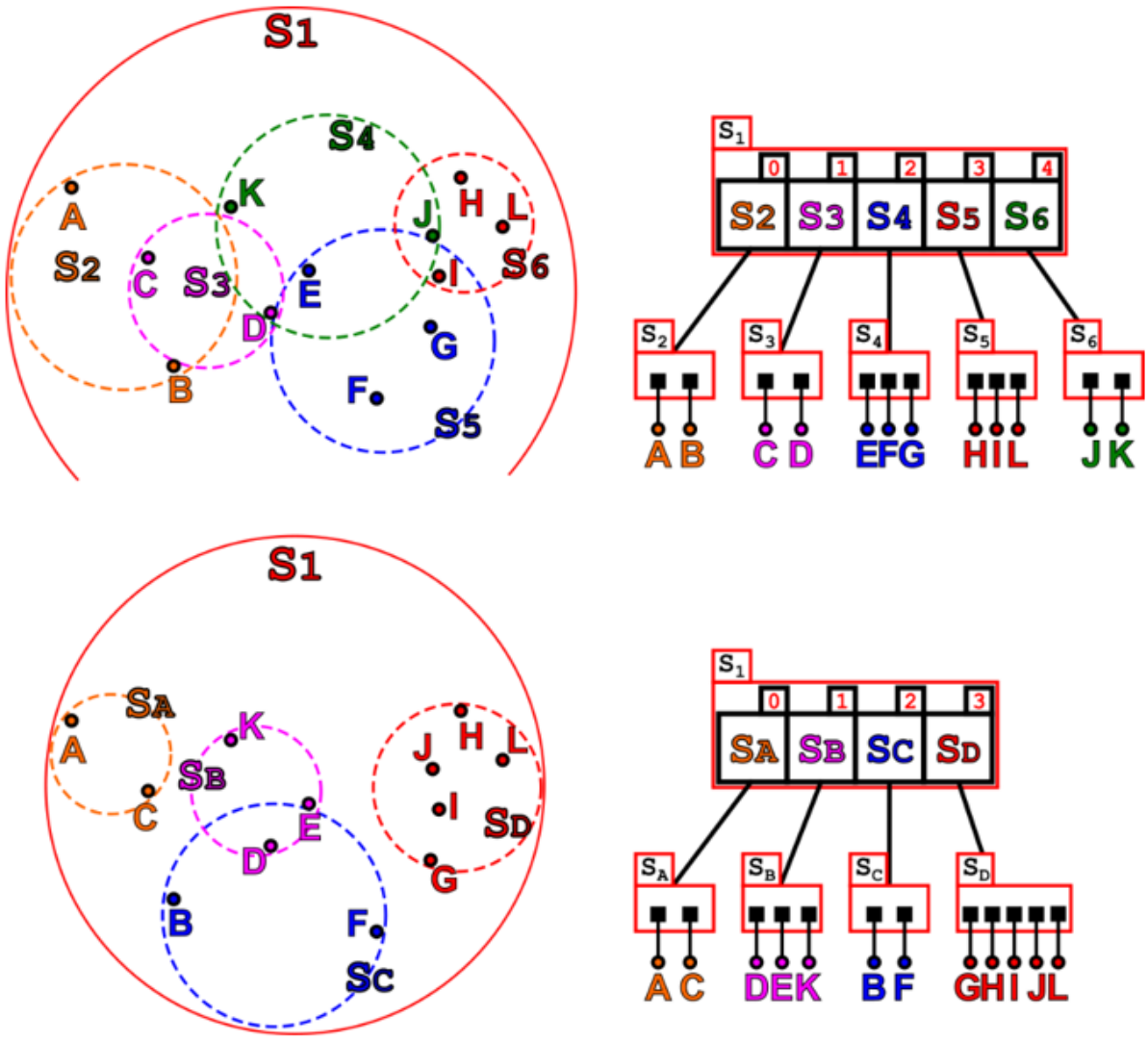


Figure 10.28 An example where the grandchildren-split heuristic could be triggered and would lower the nodes overlapping. The Ss-tree in the example has `m==2` and `M>=5`. The k-means split heuristic is run on the points `A-K`, with `k==5`. Notice how k-means can (and sometimes will) output fewer clusters than the initial number of centroids: in this case, only 4 clusters were returned.

To illustrate these situations, the top half of figure 10.28 shows several nodes, and their parents, with a significant